MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, GENERAL LIGHTING

110 WATTS, 120 VOLTS, EXPLOSIONPROOF, SYMBOLS 48.2 AND 68.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).

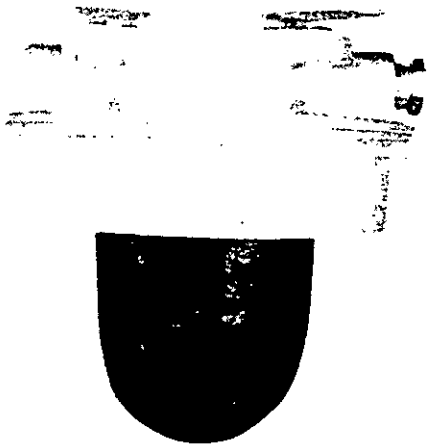


FIGURE 1. Fixtures, incandescent, Symbols 48.2 and 68.2.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

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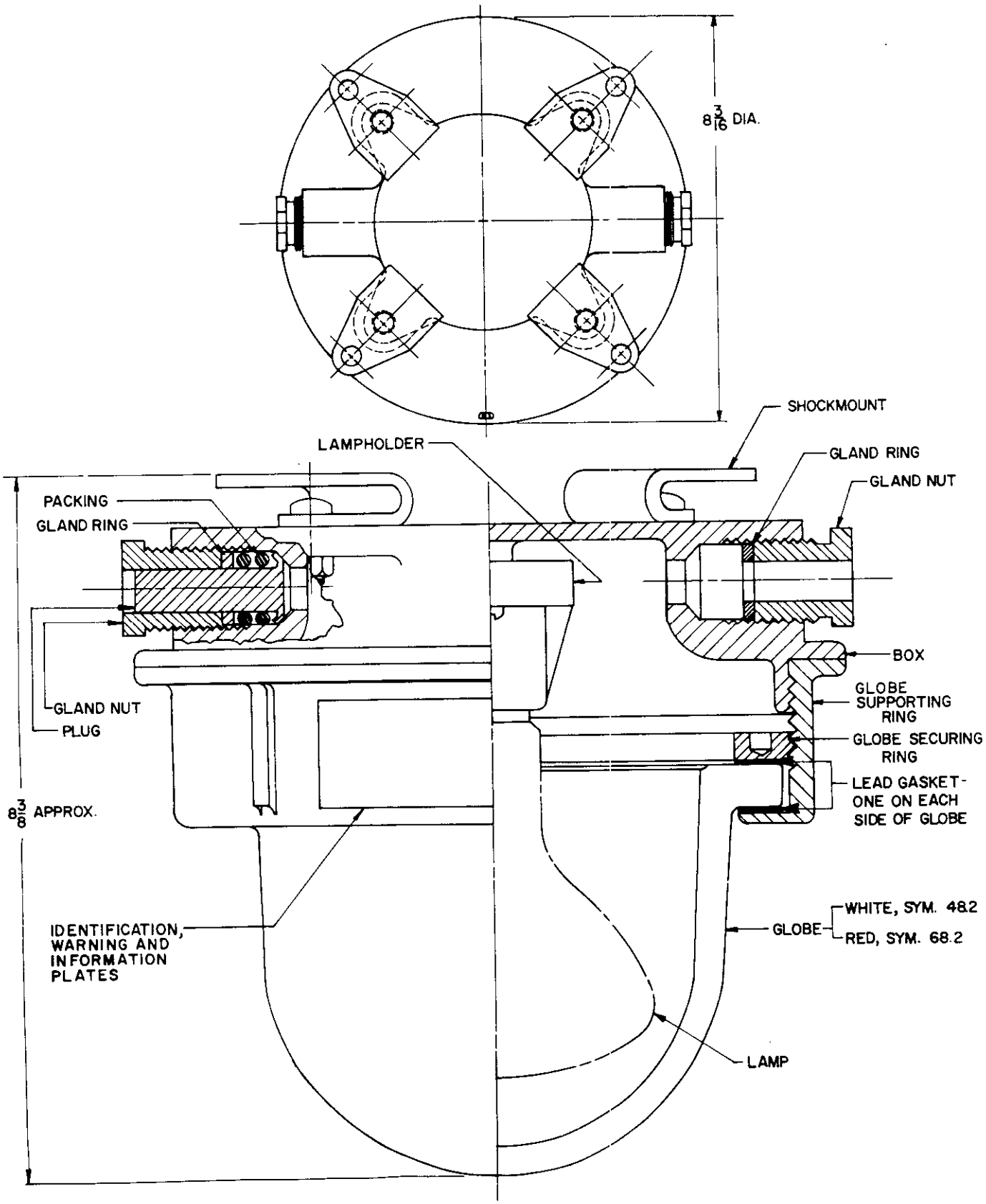


FIGURE 2. Fixture assembly.

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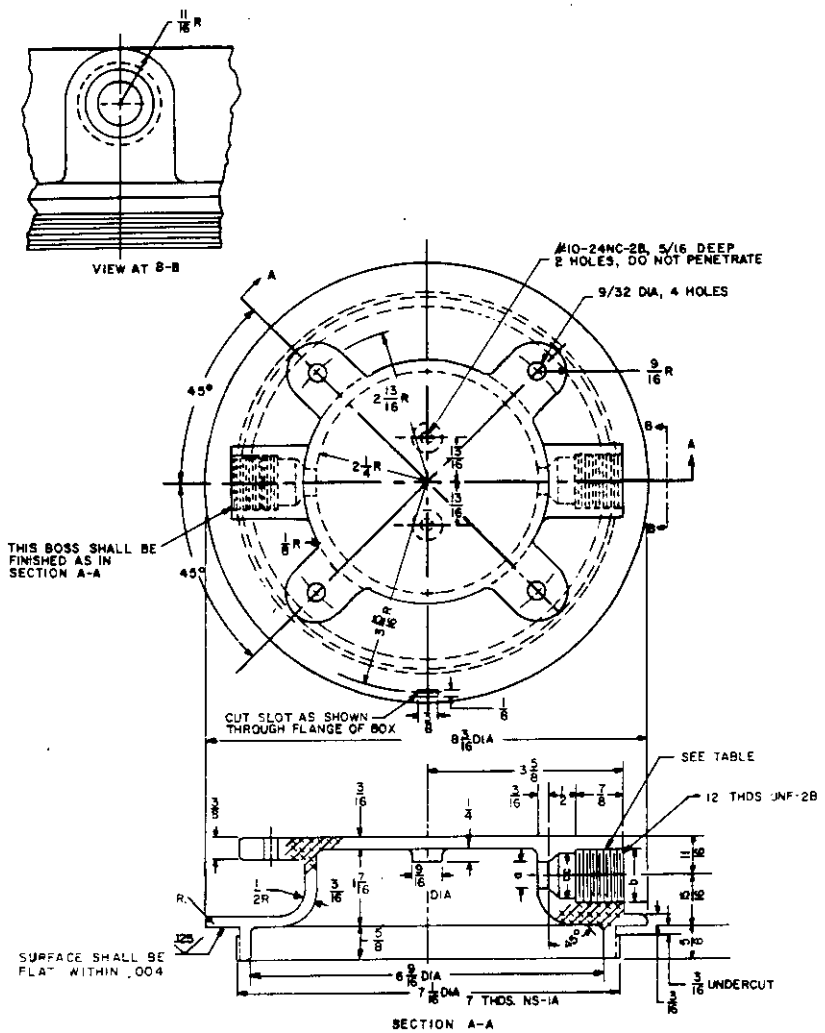


TABLE-GLAND DIMENSIONS			
SIZE	a	b	cc
"8"	".515	1"	".892

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FIGURE 3. Box.

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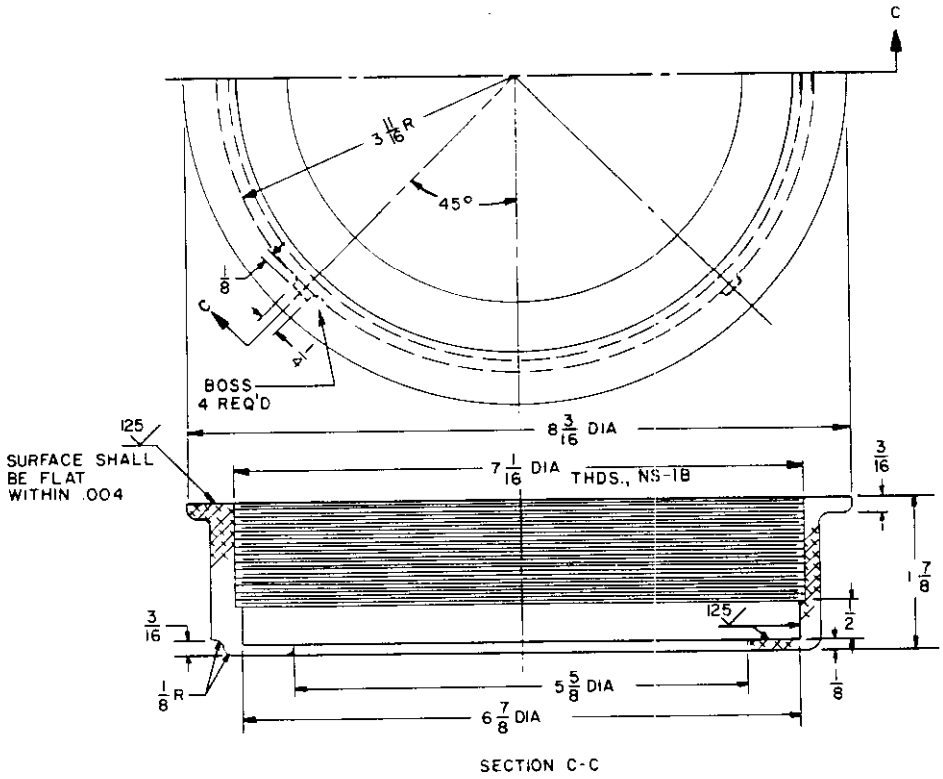


FIGURE 4. Globe supporting ring.

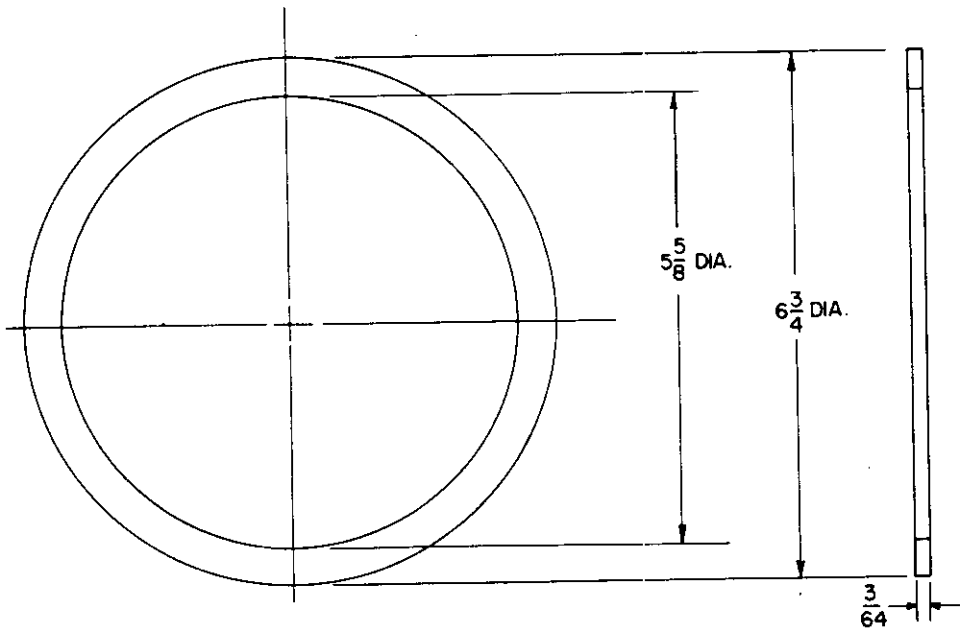
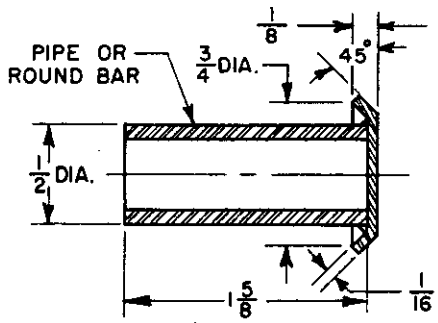


FIGURE 5. Lead gasket.

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FIGURE 6. Sealing plug.

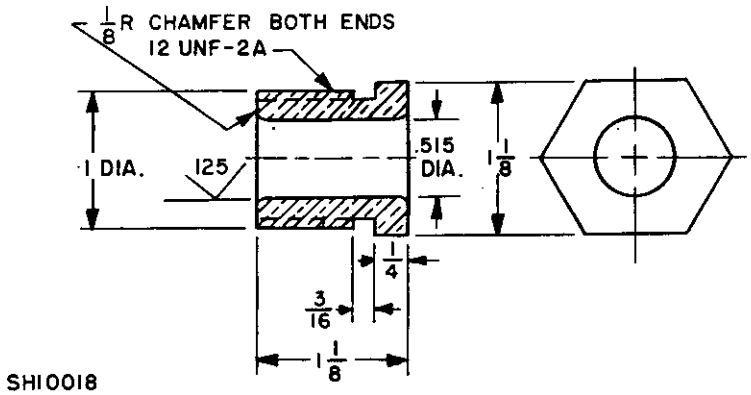


FIGURE 7. Gland nut.

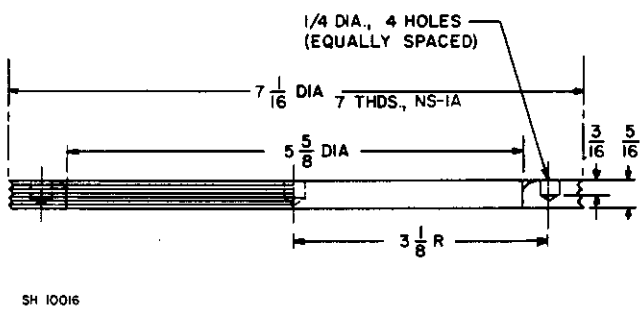
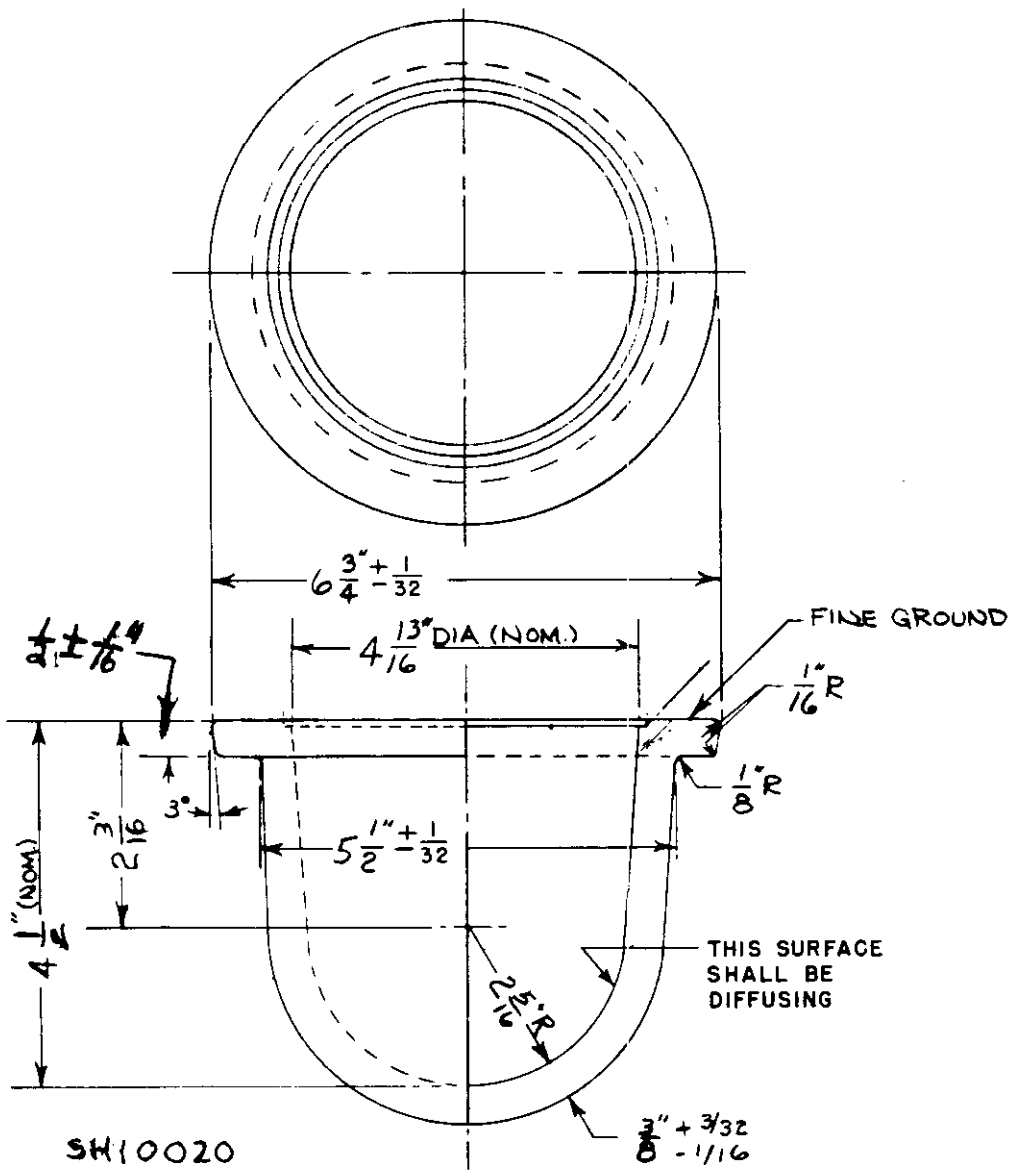


FIGURE 8. Globe securing ring.

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Material - Heat resistant, high-impact glass.
Color:
Symbol 48.2 - Clear transparent/Diffusing.
Symbol 68.2 - Red transparent/Diffusing.
Transmittance: Symbol 48.2
Clear transparent/Diffusing - 80 percent minimum.

FIGURE 9. Globe.

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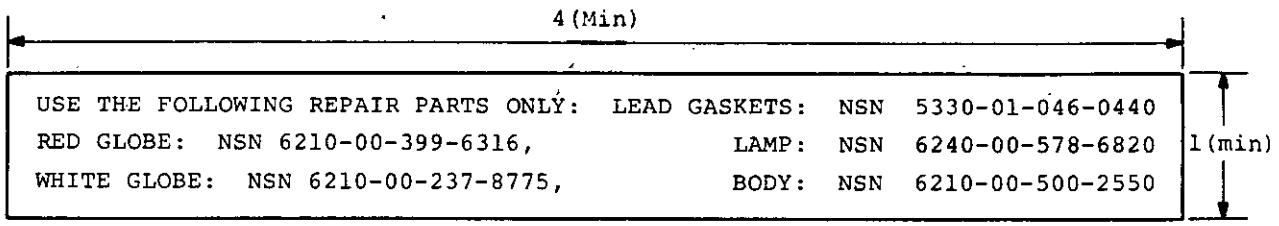


FIGURE 10. Information plate.

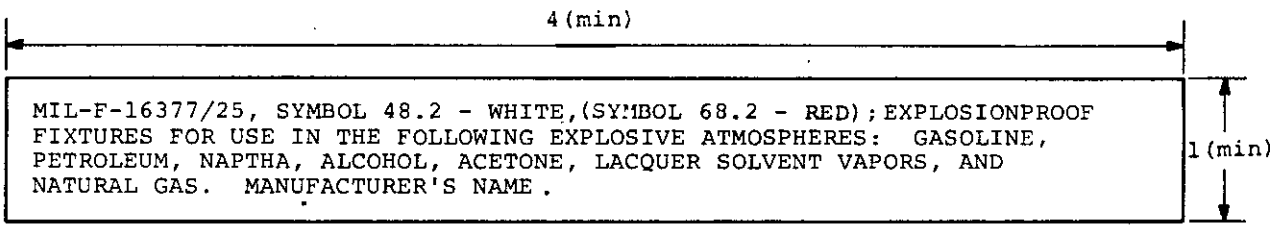


FIGURE 11. Identification plate.

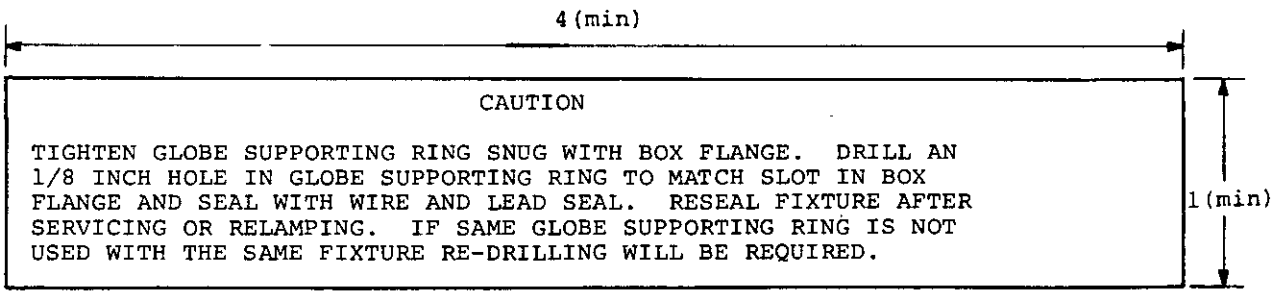
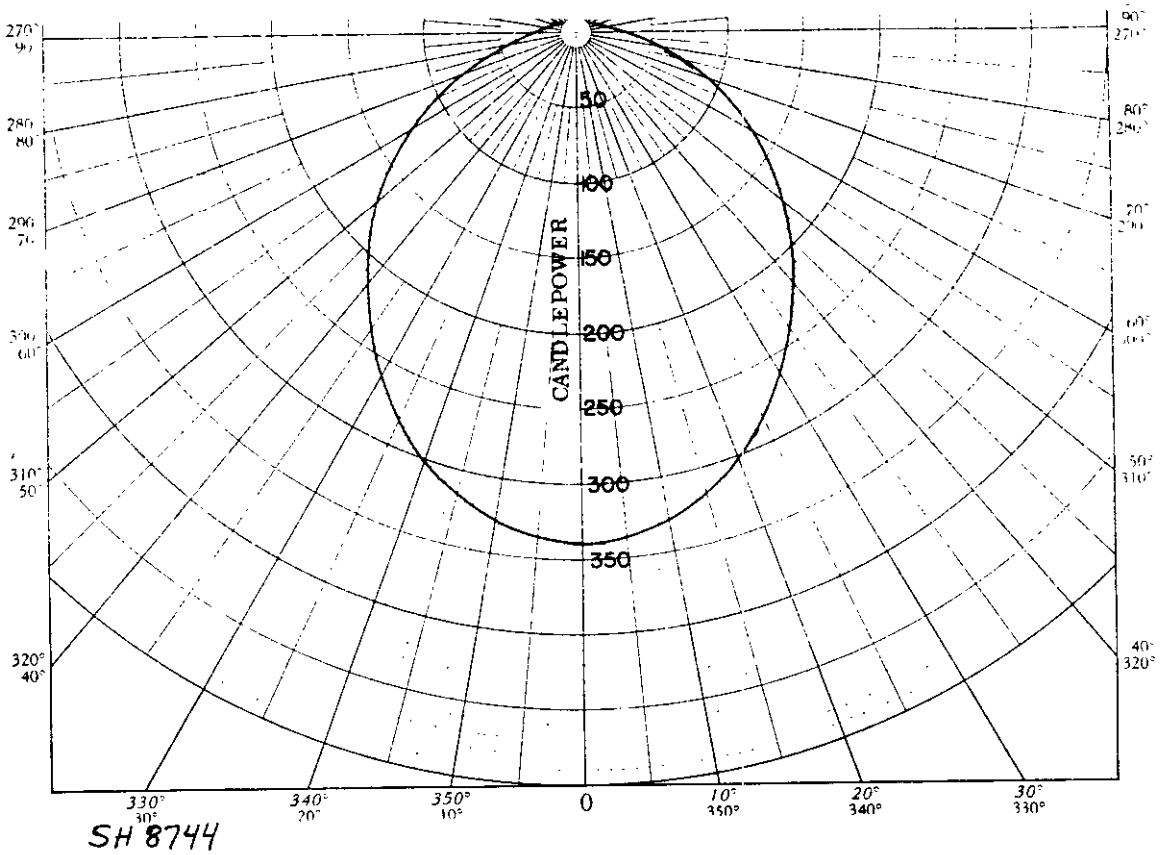


FIGURE 12. Caution plate.

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ANGLE	CANDLEPOWER SYMBOL 48.2
0	337
5	332
10	324
15	312
20	298
25	284
30	263
35	243
40	220
45	199
50	173
55	152
60	130

ANGLE	CANDLEPOWER SYMBOL 48.2
65	105
70	83
75	66
80	50
85	41
90	30
95	27
100	21
105	18
110	15
115	13
120	11
125	10

Photometric test data:

Distance: 10 feet.
Voltage: 120 volts.
Ambient temperature: 76°F.
Lamp: 110 watts, rough service conforming to MS16563-1, furnished by laboratory for test, shall be used.

FIGURE 13. Candlepower distribution curves for symbol 48.2.

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REQUIREMENTS:

CLASSIFICATION:

Type: II
Class: 2

SYMBOL NUMBERS:

48.2: Complete lighting fixture assembly with white globe.
68.2: Complete lighting fixture assembly with red globe.

DESIGN:

Electrical rating: 110 watts, 120 volts.
Dimension and configurations: See figures 1 through 12.
Photometric data: See figure 13
Globes:
Symbol 48.2: White, figure 9; Symbol 68.2: Red, figure 9
Enclosure: Explosionproof
Gland nut: Aluminum, figure 7.
Sealing plug: Aluminum, figure 6.
Gland ring: Aluminum, 7/8 o.d., 0.515 i.d. by 1/8 thick.
Box including supporting ring and globe securing ring: Aluminum, cast.
Gaskets: Lead, figure 5.
Lampholder: MIL-L-970/6, two contact.
Shockmounts: MIL-F-16377/41, part number M16377/41-006.
Weight: 7 pounds, 10 ounces max.
Lamp: These fixtures shall be designed to accommodate a 110-watt lamp conforming to MS15563-1. (Lamp not furnished.)
Designation and marking: The following plates shall be furnished in accordance with MIL-F-16377 applied as shown in figure 2. Plates shall be permanently attached to the globe supporting ring.

a. One information plate, figure 10.
b. One identification plate, figure 11.
c. One caution plate, figure 12.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I
Globes:	See table II
Quality conformance:	
Comparison inspection:	Same as the first article inspection
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

GENERAL INFORMATION:

(a) Assignment of Military part numbers and general information is shown in table III.
(b) Rope packing in accordance with Class 1, Composition A of MIL-P-16685 compressed to 1/2 minimum shall be used by installing activities for cable gland sealing. Sheet 16, Section H, Appendix B of publication NAVSEA 0964-LP-000-2000 provides additional installation information.

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TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.5	Explosion-proof
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.5.1	4.8.4	
Mechanical shock	3.7.5.6	4.8.17.3	8 ft. lbs.
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 68.2 only
Light output	3.7.1	4.8.6.1	Candlepower Symbol 48.2 only
Magnetic permeability	3.4.8	4.8.18	

TABLE II. First article and comparison inspection for globes.

Test	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.5.1	4.8.4	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Light output	3.7.1	4.8.6.1	Candlepower Symbol 48.2 only
Mechanical shock	3.7.5.6	4.8.17.3	8 ft. lbs.
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 68.2 only

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TABLE III. Part number and general information.

Military Part Number M16377/25-	Item Name	National Stock Number (NSN)	Remarks
48.2	Complete fixture, Symbol 48.2 Consisting of: Body White Globe	 6210-00-500-2550 6210-00-237-8775	Figure 2, 1/
68.2	Complete fixture, Symbol 68.2 Consisting of: Body Red transparent diffusing globe	6210-00-500-2550 6210-00-399-6316	Figure 2, 1/
001	Body - Complete fixture less globe	6210-00-500-2550	Figure 2
002	White Globe	6210-00-174-4333	2/
003	Red transparent diffusing globe	6210-00-399-6316	3/
004	Lead gaskets - a set of 2 gaskets	5330-01-046-0440	Figure 5
005	Globe sealing ring		Figure 8
006	Globe supporting ring		Figure 4

1/ Formerly covered on NAVSHIPS Drawing 9-S-5021-L and MS17308.
2/ Formerly covered on MS17802, part MS17802-13.
3/ Formerly covered on MS17802, part MS17802-14

Preparing activity:
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